

STATE TRANSITION TESTING

A Case Study on Flipkart, Naukri, Facebook, Myntra and Swiggy

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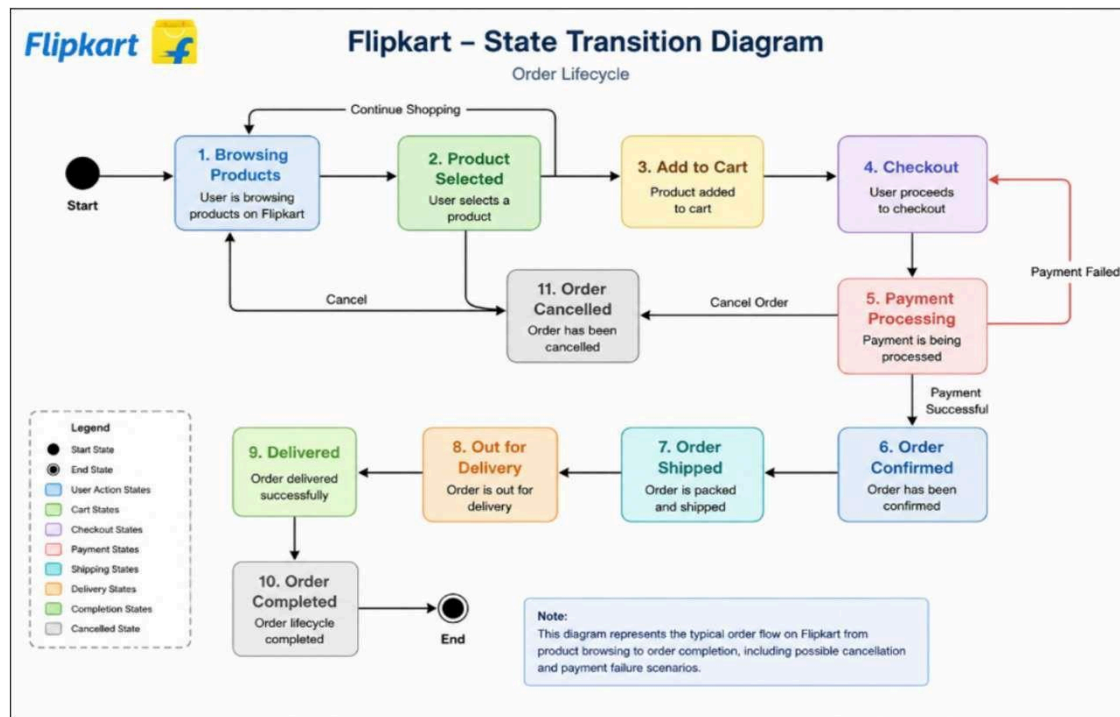
Introduction

State transition testing is a black-box technique that checks whether a system moves correctly from one state to the next in response to user actions or events. Rather than looking at inputs and outputs in isolation, this method maps out every state the application can be in and confirms that both the expected and the unexpected transitions are handled the way they should be.

Objective

- Confirm that the application correctly moves from one state to the next.
- Check that incorrect or unexpected transitions are handled gracefully.
- Trace the complete workflow a user would follow from start to finish.
- Strengthen overall confidence in the reliability of the application.

1. Flipkart – State Transition Flow



States Covered

- Browsing Products
- Product Selected
- Item Added to Cart
- Checkout
- Payment in Progress
- Order Confirmed
- Order Shipped
- Out for Delivery
- Delivered
- Order Completed
- Order Cancelled

Sample Test Cases

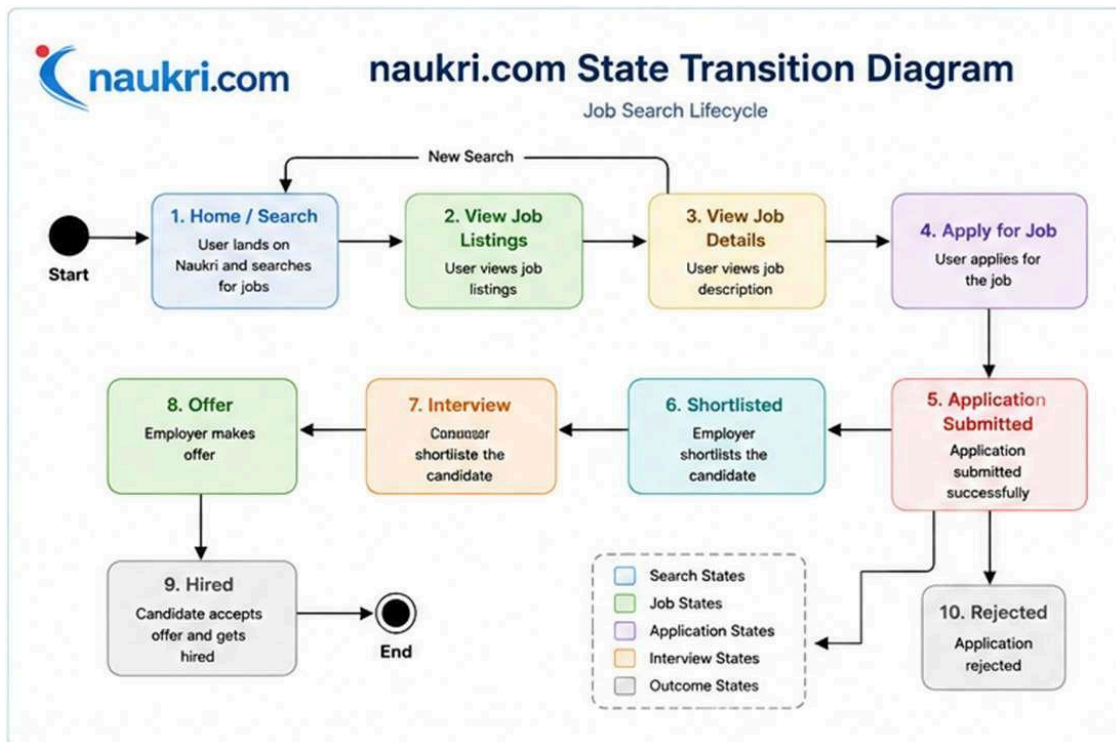
Test Case ID	Starting State	Triggering Action	Resulting State
ST1_01	Browsing Products	User selects a product	Product Selected
ST1_02	Product Selected	User adds item to cart	Item Added to Cart
ST1_03	Checkout	Payment goes through	Order Confirmed
ST1_04	Checkout	Payment does not go through	Payment Failed
ST1_05	Order Confirmed	Seller ships the order	Order Shipped

Test Case ID	Starting State	Triggering Action	Resulting State
ST1_06	Order Shipped	Order handed to delivery agent	Out for Delivery
ST1_07	Out for Delivery	Package reaches the customer	Delivered
ST1_08	Payment in Progress	User cancels the order	Order Cancelled

Explanation

These scenarios check that a Flipkart order moves through each stage correctly, and that the app reacts sensibly when a payment does not succeed or when the customer cancels midway.

2. Naukri – State Transition Flow



States Covered

- Home / Job Search
- Job Listings
- Job Details
- Application Submitted
- Applied
- Shortlisted
- Interview
- Offer
- Hired
- Rejected

Sample Test Cases

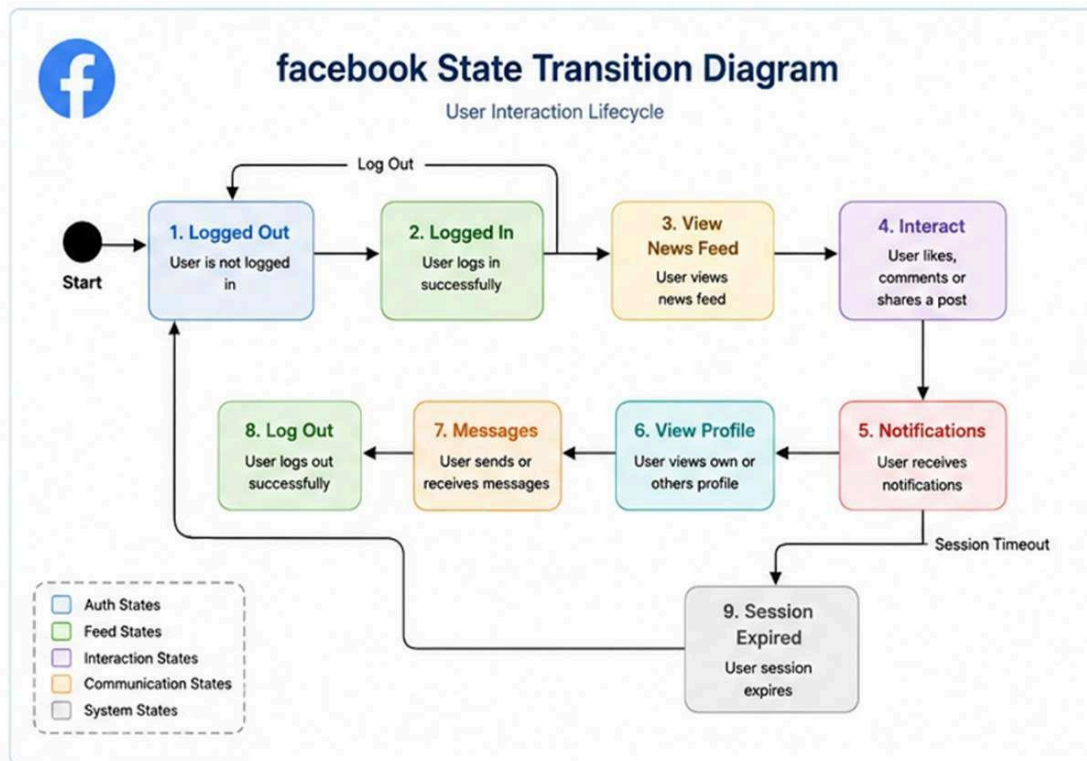
Test Case ID	Starting State	Triggering Action	Resulting State
ST2_01	Home	Candidate searches for a job	Job Listings
ST2_02	Job Listings	Candidate opens a listing	Job Details
ST2_03	Job Details	Candidate clicks Apply	Application Submitted
ST2_04	Application Submitted	Recruiter shortlists the candidate	Shortlisted

Test Case ID	Starting State	Triggering Action	Resulting State
ST2_05	Shortlisted	Interview round is completed	Offer
ST2_06	Application Submitted	Recruiter turns down the application	Rejected

Explanation

These scenarios confirm that a candidate's journey – from searching a job to receiving an outcome – follows the correct order and lands on the right result at each step.

3. Facebook – State Transition Flow



States Covered

- Logged Out
- Logged In
- News Feed
- Post Interaction
- Notifications
- Profile View
- Messages
- Logged Out (via Log Out)
- Session Expired

Sample Test Cases

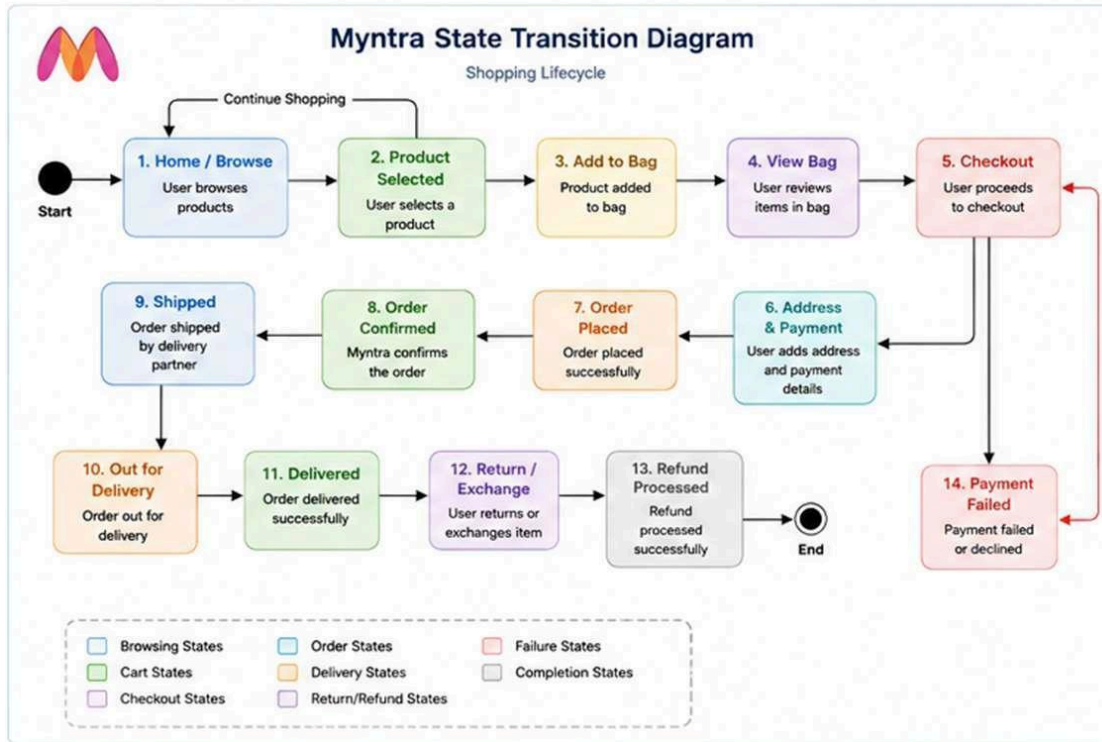
Test Case ID	Starting State	Triggering Action	Resulting State
ST3_01	Logged Out	User signs in	Logged In
ST3_02	Logged In	User opens the feed	News Feed
ST3_03	News Feed	User likes or comments on a post	Post Interaction
ST3_04	Post Interaction	A notification is generated	Notifications
ST3_05	Logged In	User signs out	Logged Out

Test Case ID	Starting State	Triggering Action	Resulting State
ST3_06	Logged In	Session times out	Session Expired

Explanation

These scenarios make sure a user can move between the main screens without issue, and that the app handles sign-out and session expiry the way it is expected to.

4. Myntra – State Transition Flow



States Covered

- Browsing Products
- Product Selected
- Added to Bag
- Bag Review
- Checkout
- Payment
- Order Placed
- Order Confirmed
- Shipped
- Out for Delivery
- Delivered
- Return / Exchange Requested
- Refund Processed
- Payment Failed

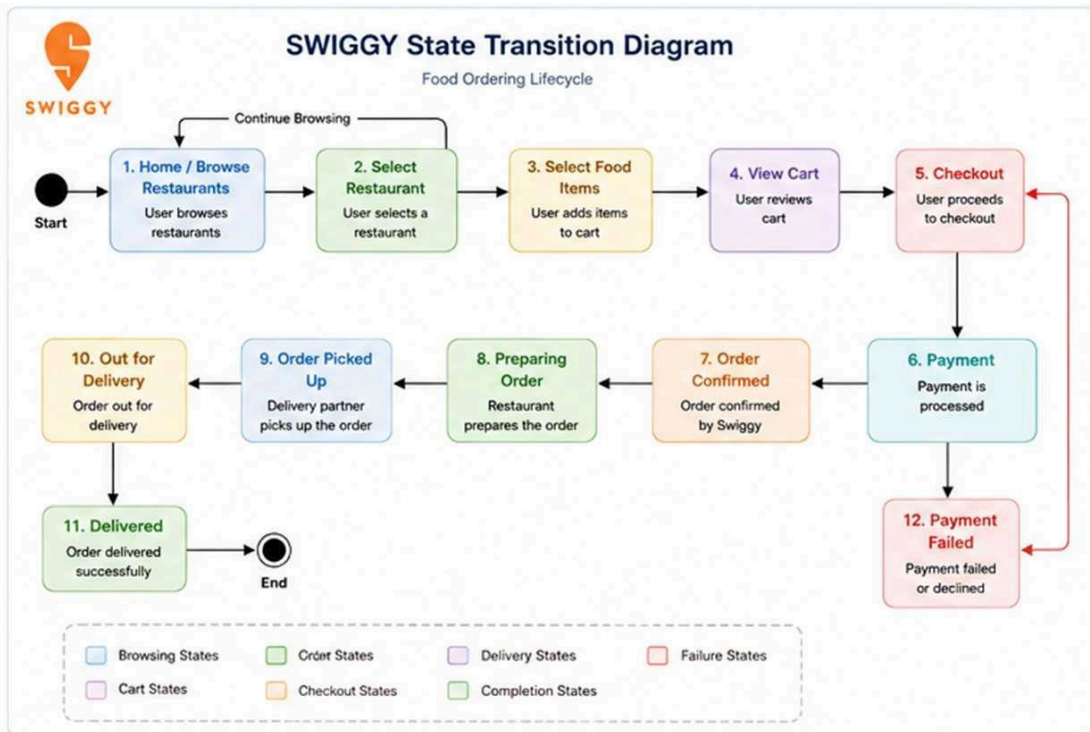
Sample Test Cases

Test Case ID	Starting State	Triggering Action	Resulting State
ST4_01	Browsing	User picks a product	Product Selected
ST4_02	Product Selected	User adds product to bag	Added to Bag
ST4_03	Checkout	Payment succeeds	Order Placed
ST4_04	Checkout	Payment does not succeed	Payment Failed
ST4_05	Delivered	Customer raises a return request	Return / Exchange Requested
ST4_06	Return Approved	Refund is initiated	Refund Processed

Explanation

These scenarios cover the regular shopping path as well as the after-sales flow, verifying both order placement and the return-and-refund process behave as intended.

5. Swiggy – State Transition Flow



States Covered

- Browsing Restaurants
- Restaurant Selected
- Food Items Selected
- Cart Review
- Checkout
- Payment
- Order Confirmed
- Order Being Prepared
- Order Picked Up
- Out for Delivery
- Delivered
- Payment Failed

Sample Test Cases

Test Case ID	Starting State	Triggering Action	Resulting State
ST5_01	Browsing Restaurants	User picks a restaurant	Restaurant Selected
ST5_02	Selecting Food	User adds items to the cart	Cart Review
ST5_03	Checkout	Payment succeeds	Order Confirmed

Test Case ID	Starting State	Triggering Action	Resulting State
ST5_04	Checkout	Payment does not succeed	Payment Failed
ST5_05	Order Being Prepared	Delivery partner collects the order	Order Picked Up
ST5_06	Out for Delivery	Order reaches the customer	Delivered

Explanation

These scenarios confirm that a food order advances correctly through each stage of the Swiggy ordering process, including proper handling of failed payments.